

V3 Pipeline — Egitim Raporu

Tarih: 2026-03-22 20:31

PIPELINE OZELLIKLERI:

Feature Engineering: V3 (Label-encoded predictions + ayri OHE + manuel k-mer)
Feature sayisi: 989
Toplam ornek: 4287 (CV: 3429, Hold-Out: 858)
Model turleri: LightGBM, XGBoost
Hyperparameter: Grid Search (12 combo/model, 3-fold CV)
Degerlendirme: 5-Fold Stratified CV + %20 Hold-Out Test

5-FOLD CV SONUCLARI (egitim seti %80):

lightgbm : F1 = 0.9841 +/- 0.0015
xgboost : F1 = 0.9850 +/- 0.0014

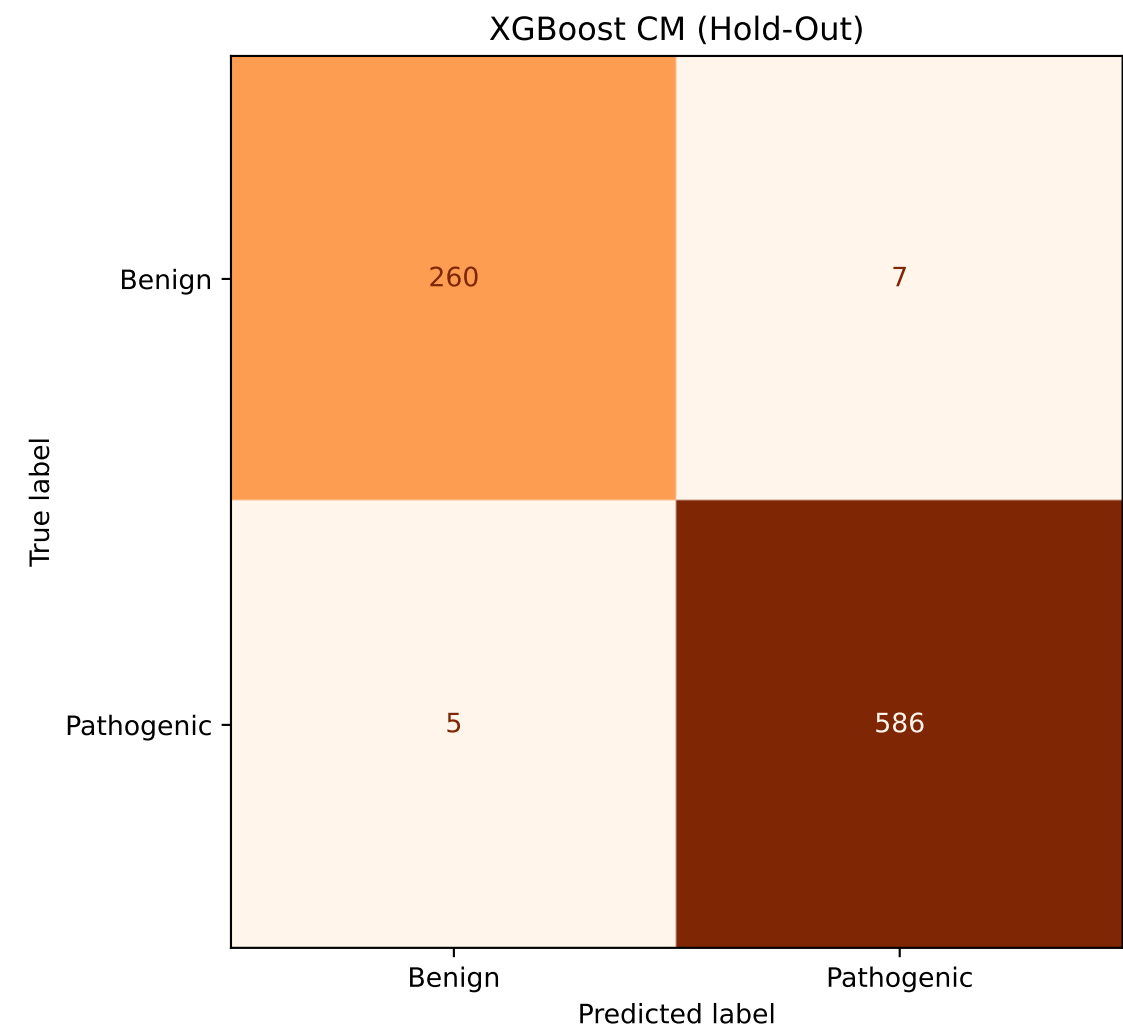
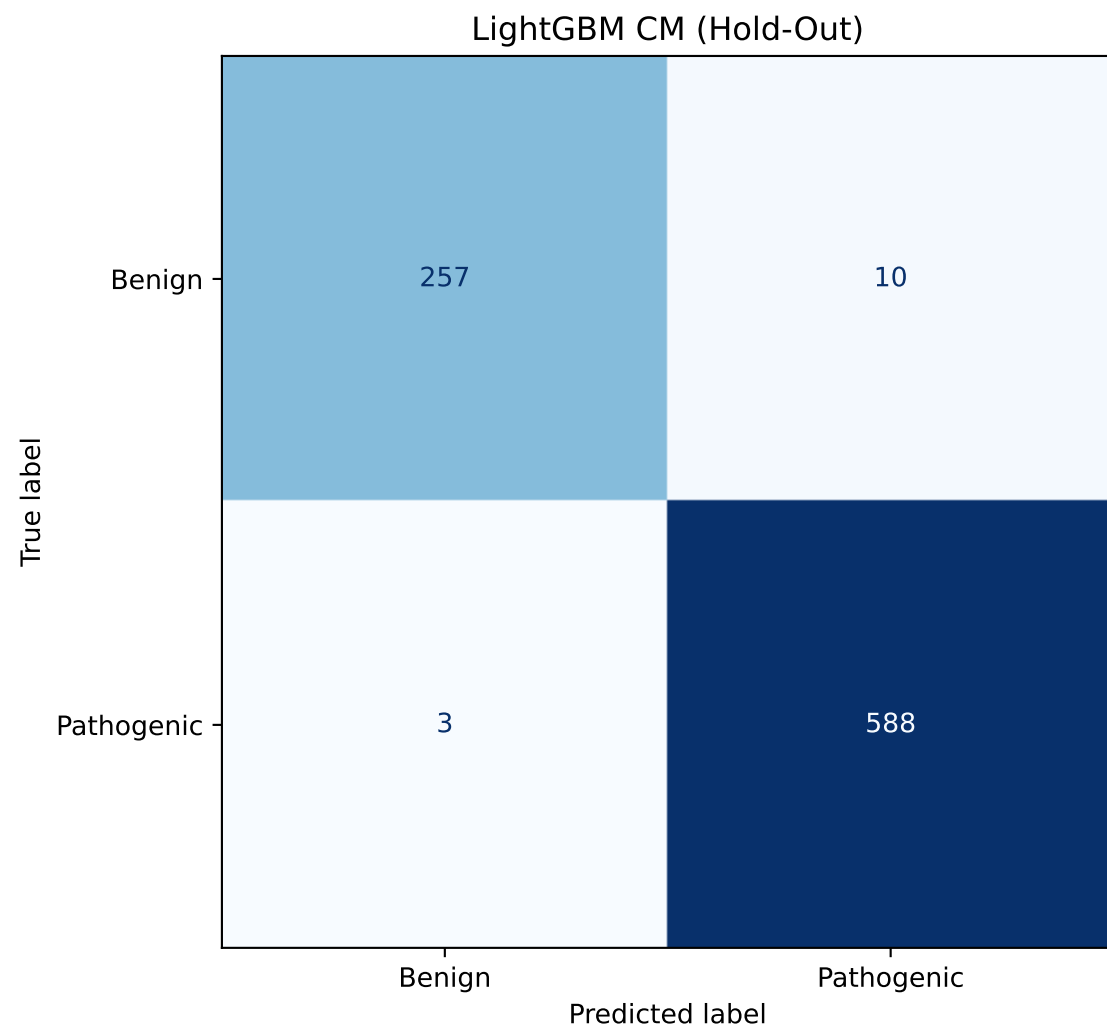
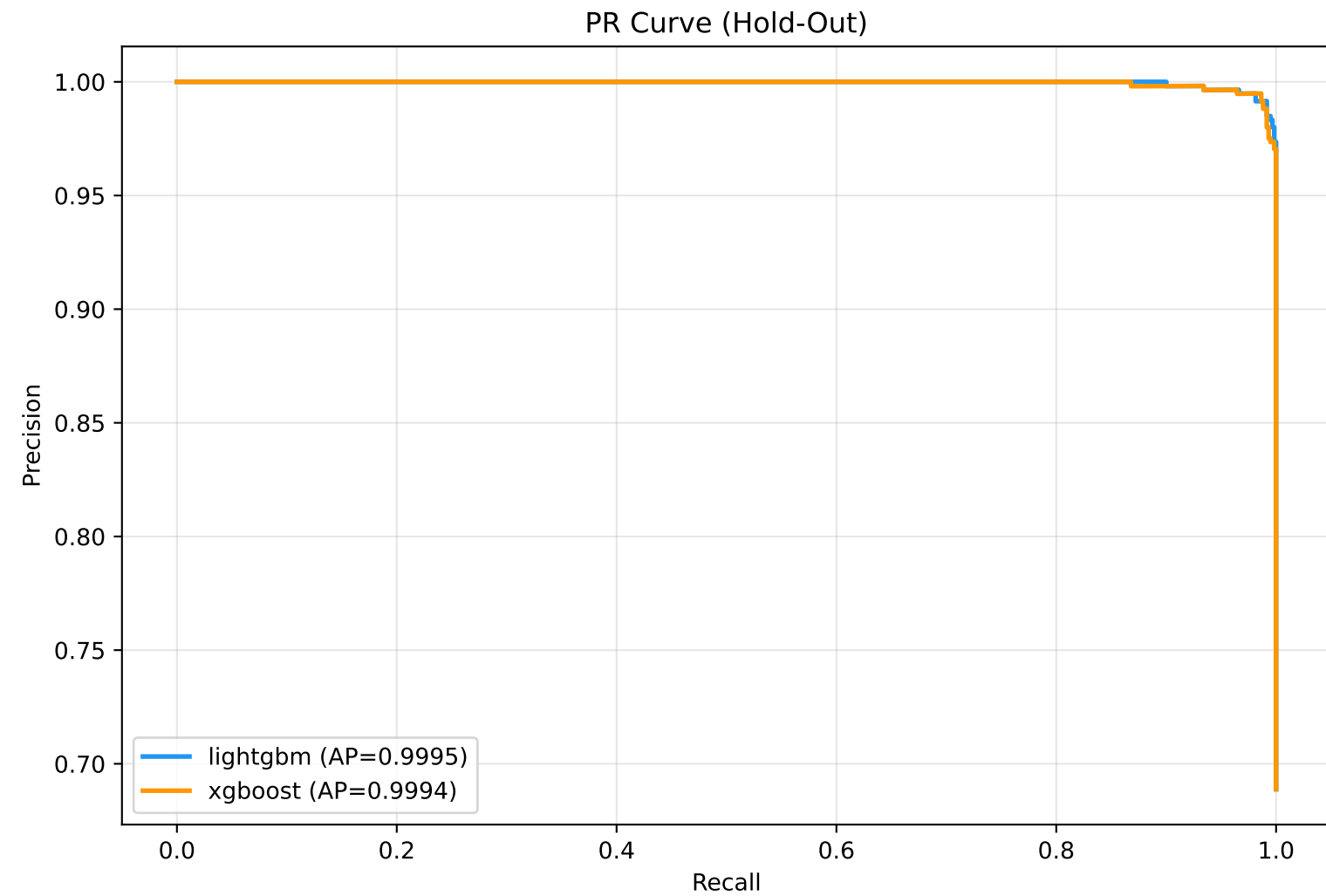
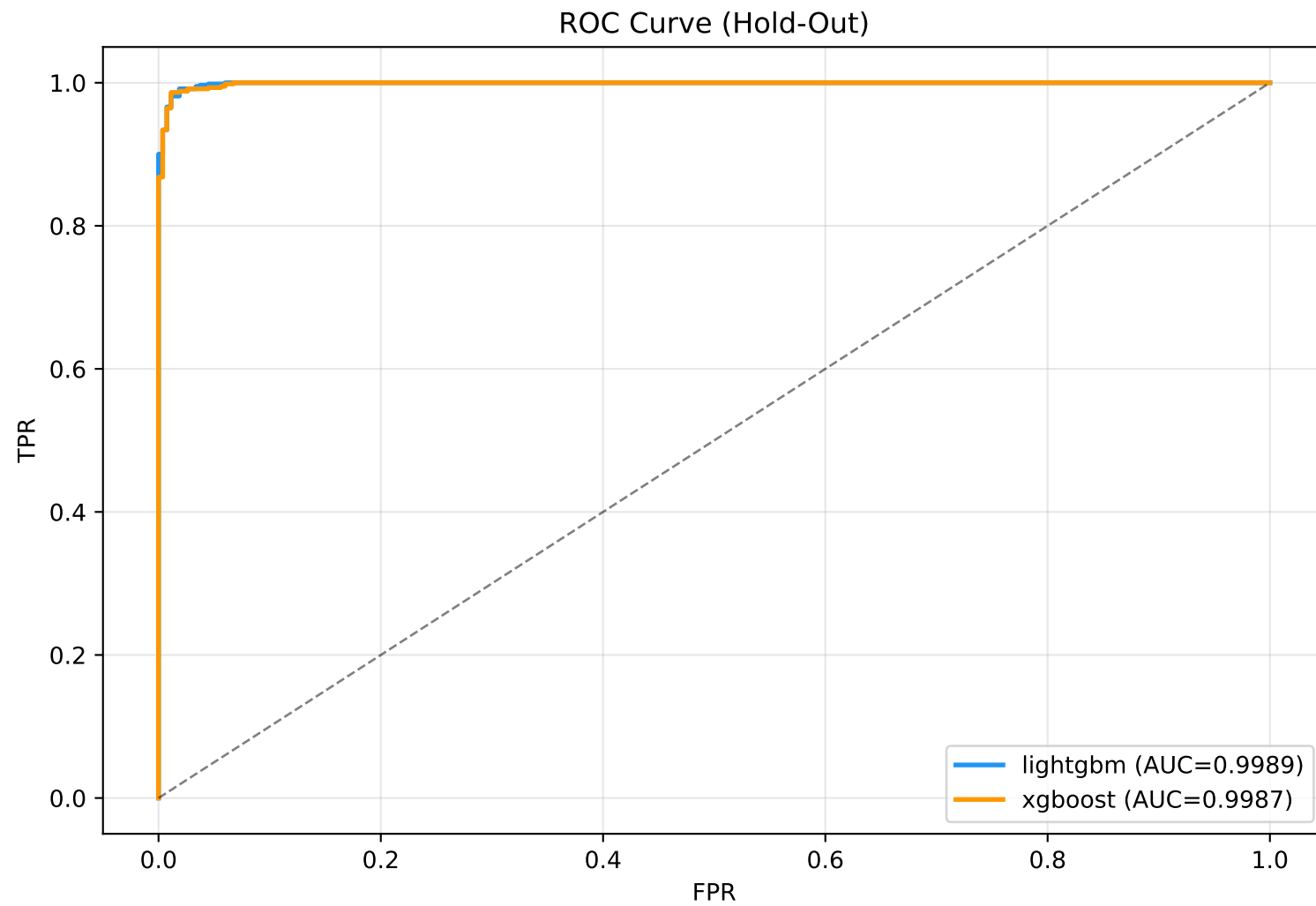
HOLD-OUT TEST SONUCLARI (hic gorulmemis %20):

lightgbm : F1=0.9891 AUC=0.9989 (CV F1=0.9841)
xgboost : F1=0.9899 AUC=0.9987 (CV F1=0.9850)

KARSILASTIRMA (full_fe vs v3 hold-out):

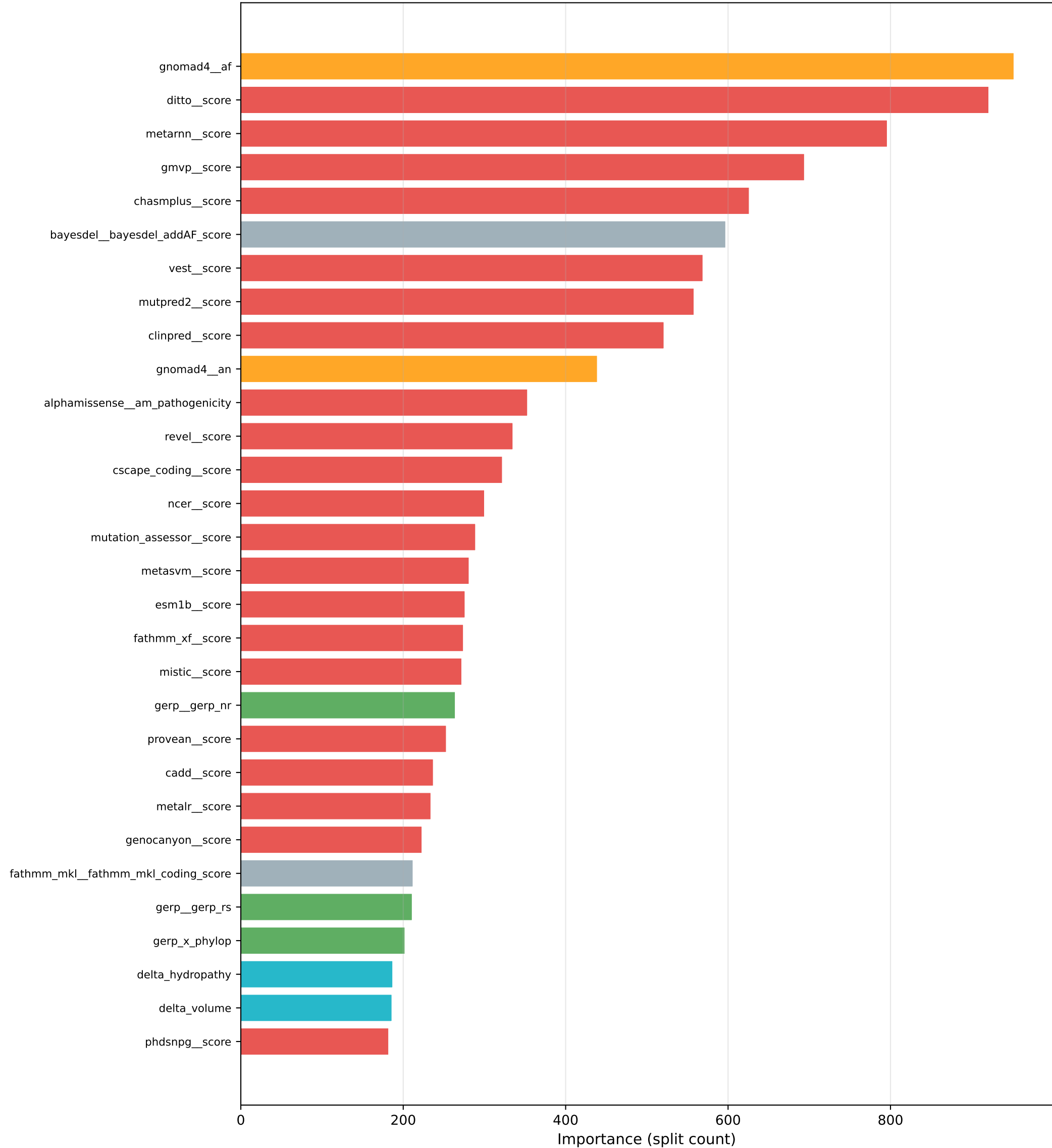
lightgbm: full_fe F1=0.6967 -> v3 Hold-Out F1=0.9891 (+0.2924)
xgboost: full_fe F1=0.6879 -> v3 Hold-Out F1=0.9899 (+0.3020)

Hold-Out Test Sonuclari (%20 — Hic Gorulmemis Veri)

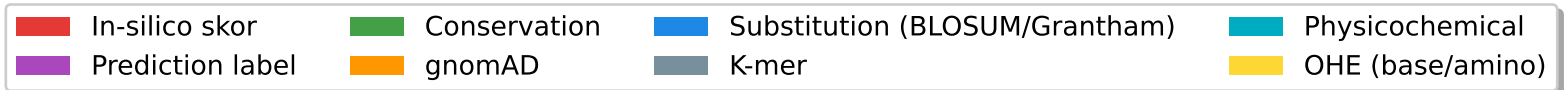
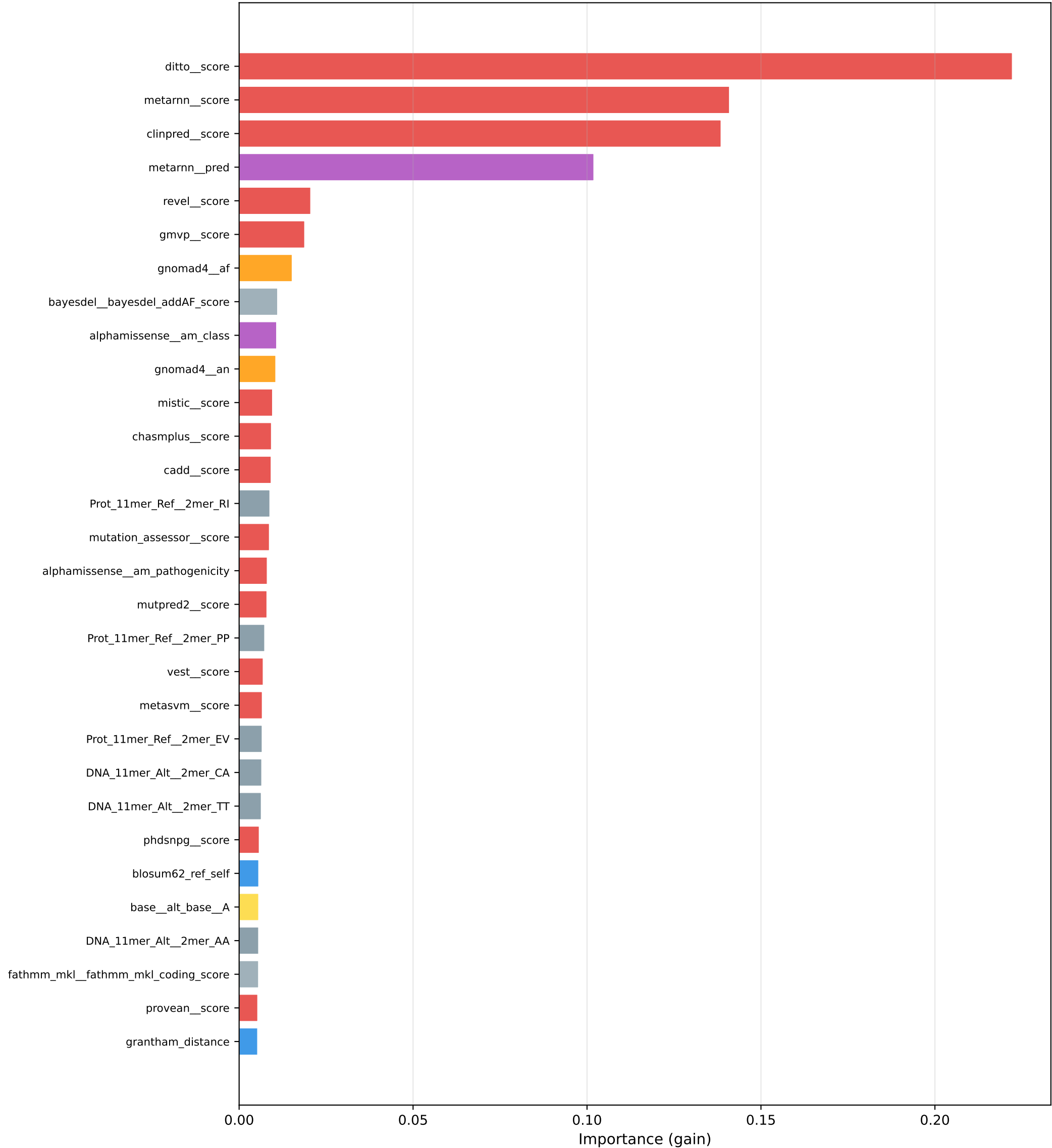


Feature Importance — Top 30 (En İyi CV Fold Modeli)

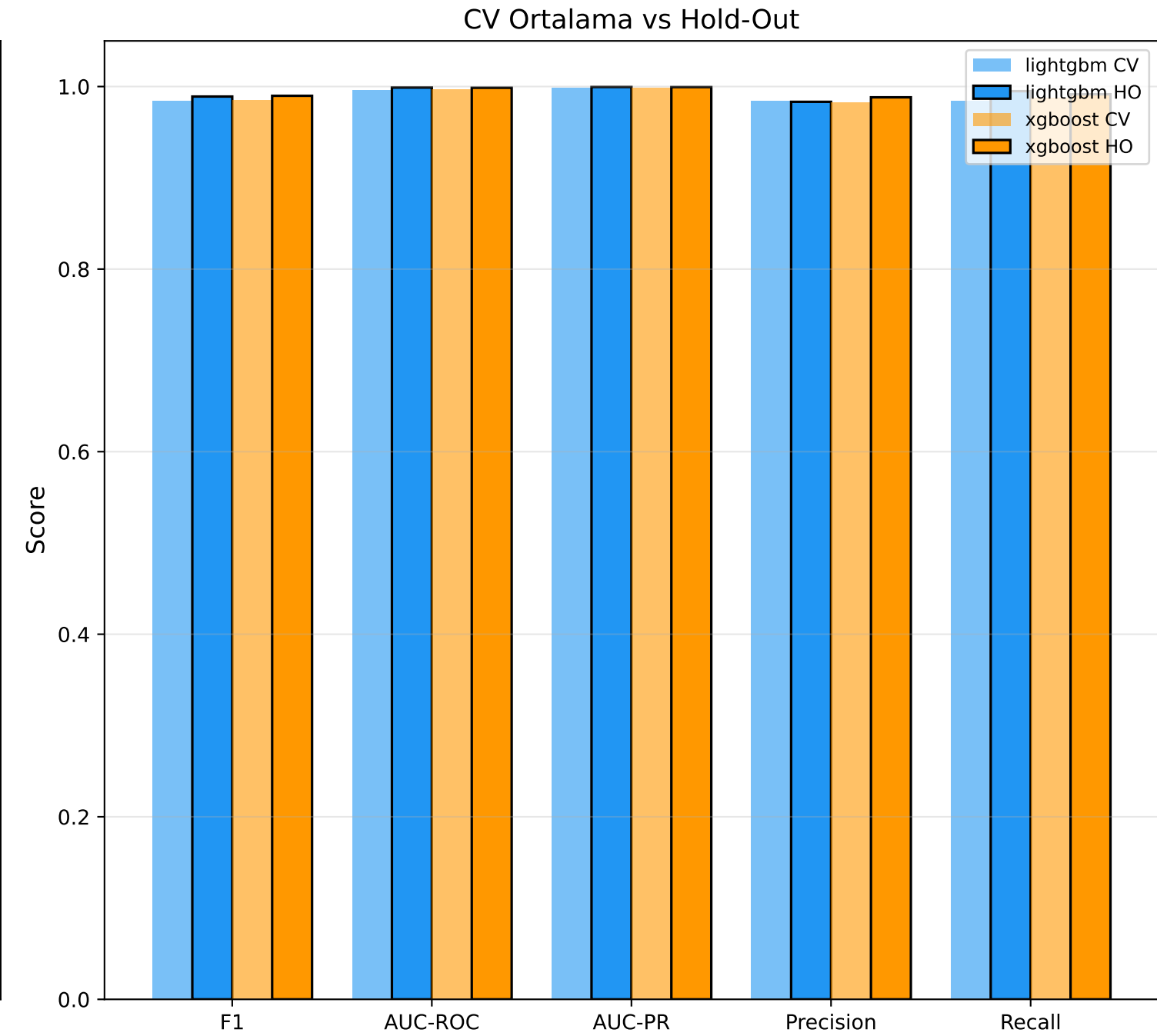
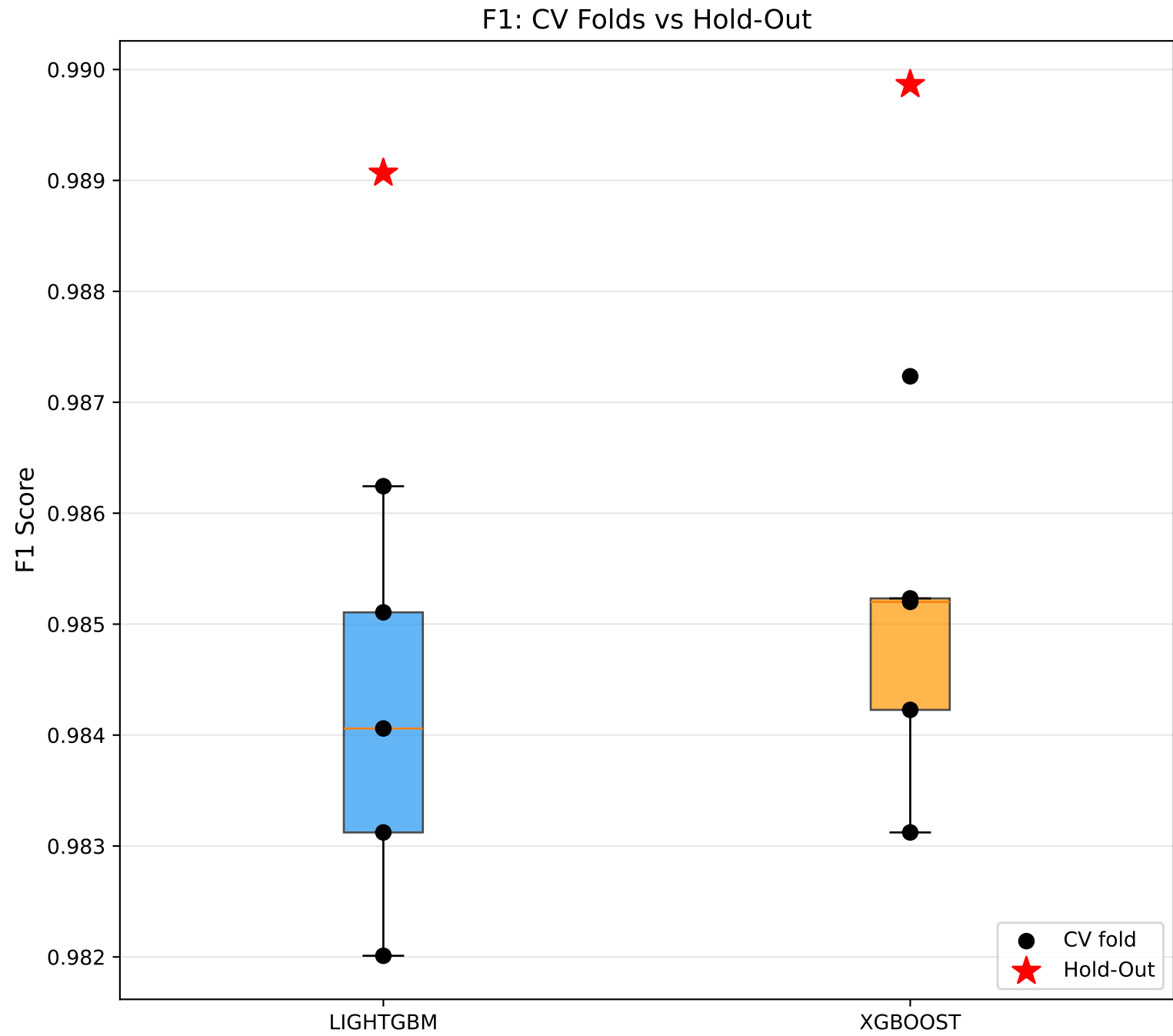
LightGBM



XGBoost



CV vs Hold-Out Karsilastirma



Detayli Sonuclar: 5-Fold CV + Hold-Out Test

Fold	Model	Split	F1	AUC-ROC	AUC-PR	Precision	Recall	MCC
1	lightgbm	CV	0.9862	0.9950	0.9981	0.9852	0.9873	0.9558
2	lightgbm	CV	0.9820	0.9962	0.9982	0.9810	0.9831	0.9422
3	lightgbm	CV	0.9831	0.9958	0.9978	0.9790	0.9873	0.9455
4	lightgbm	CV	0.9851	0.9980	0.9991	0.9893	0.9809	0.9528
5	lightgbm	CV	0.9841	0.9968	0.9985	0.9872	0.9809	0.9491
1	xgboost	CV	0.9852	0.9958	0.9983	0.9811	0.9894	0.9523
2	xgboost	CV	0.9852	0.9960	0.9982	0.9831	0.9873	0.9524
3	xgboost	CV	0.9831	0.9964	0.9983	0.9790	0.9873	0.9455
4	xgboost	CV	0.9872	0.9980	0.9991	0.9915	0.9831	0.9595
5	xgboost	CV	0.9842	0.9974	0.9988	0.9770	0.9915	0.9487
ORT	lightgbm	CV	0.9841	0.9964	0.9983	0.9843	0.9839	0.9491
ORT	xgboost	CV	0.9850	0.9967	0.9985	0.9823	0.9877	0.9517
-	lightgbm	HOLD-OUT	0.9891	0.9989	0.9995	0.9833	0.9949	0.9646
-	xgboost	HOLD-OUT	0.9899	0.9987	0.9994	0.9882	0.9915	0.9673